Two reductions for Diophantine consistency

A two-root equation of degree 278 and a smaller direct integer quartic

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Abstract

We give two explicit reductions relative to the ordinary calibrated source theorems in a fixed ProveIt snapshot. The first gives an integer-coefficient polynomial in ten strictly positive unknowns with exact degree 278 and a 93-operation evaluator. Its immediate predecessor has degree 368 and 99 operations. The construction saturates a strong Pell discriminant by the first Pell radicand, recovers two integral roots from their coprime product, and uses an upper polynomial center to shorten the transport condition. Two exact center identities establish the degree.

The second changes the shared numerical bound in the direct proof-certificate construction. Squaring selected signed outputs permits recovery of their intended powers before imposing their signs. Together with additional exact source consequences this yields a quartic in 576 unrestricted integers with 7,172 terms; a smaller-coefficient companion has 582 integers and 6,200 terms. These improve the corresponding 604- and 609-variable source equations. Every coefficient, residual, arithmetic circuit and reconstruction certificate is supplied in the packet.

Both reductions preserve existence of a finite ZFC refutation. Composing the ordinary source calibrations gives ZFC-provable equivalences between unsolvability and syntactic Con(ZFC). The new algebra is checked exactly, but no new end-to-end Lean/Rocq or encoded object-ZFC proof is claimed. Neither global optimality nor the consistency of ZFC is asserted.

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1 Results, baseline, and meaning of simplicity

All source comparisons use ProveIt commit `84e16e43092614bae262ca59651cb9d77e31a255`. This is essential: the active research is considerably ahead of the numbered manuscript collection. The 99-operation zero-offset equation and the 75-predicate direct quartic family are the immediate baselines, not the older 101-operation printed page or earlier hundreds-of-operation norm constructions [1, 5].

Construction	Domain	Unknowns	Degree	Operations
Zero-offset parent	$\mathbb{Z}_{>0}$	10	368	99
Two-root equation, this packet	$\mathbb{Z}_{>0}$	10	278	93
Simple integer substitution	$\mathbb{Z}$	30	556	163

The first improvement saves six live arithmetic operations and lowers degree by 90, or about 24.5%, without adding a positive unknown. Operations are binary additions, subtractions and multiplications, including squaring and multiplication by a small numeral. Named polynomial abbreviations are not additional unknowns. Fixed coefficient construction is accounted for separately.

Unrestricted integer quartic	Unknowns	Terms	Coefficient height
Matched prior small-variable profile	604	9,676	2,445,790,974,704
New small-variable profile	576	7,172	2,445,790,974,704
Matched prior smaller-height profile	609	8,823	964,213
New smaller-height profile	582	6,200	964,213

Here a term means a nonzero collected monomial, and height is the maximum absolute coefficient of the complete expanded polynomial. Thus the first quartic saves 28 variables and 2,504 terms at the same height. The second saves 27 variables and 2,623 terms at the same height. They are two measured tradeoffs, not a claim to have characterized a complete Pareto frontier.

There is no established scalar ordering that makes all these objects "simplest." A quartic in many integer variables and a degree-278 polynomial in ten positive variables answer different questions. The repository also retains fewer-variable integer constructions based on Sun's reduction, and nine-positive constructions with much larger degree. Sun's general ten-integer theorem likewise is not displaced by a ten-positive result [7]. The present contributions are concrete improvements on two specified routes.

1.1 What is proved inside ZFC

Fix the syntactic ZFC proof predicate used by the repository and write Con(ZFC) for the assertion that it has no finite refutation. The desired statement for an integer polynomial P and a domain D is

$$\mathrm{ZFC} \vdash \left(\forall \mathbf{x} \in D^m \; P(\mathbf{x}) \neq 0\right) \longleftrightarrow \mathrm{Con(ZFC)}.$$

This is an equivalence, not a proof of either side. It does not contradict the second incompleteness theorem. We give ordinary mathematical derivations of the new arithmetic equivalences and explicitly name the inherited source calibrations. The finite algorithms used in the proofs, including recurrence periods, are justified by ordinary finite arithmetic and finite group arguments within ZFC; no standard-model-only soundness premise is added.

Three evidence levels should remain distinct. The new polynomial identities and all delivered coefficients have exact executable checks. The unbounded soundness and completeness arguments are ordinary proofs.

The repository's native/calibration and internal-ZFC formalization boundaries remain as documented in the source snapshot. This packet does not close those formal development tasks merely by running a polynomial checker.

2 The complete ten-positive-variable equation

The only unknowns are

$$J, F, \alpha, i, T, w, Y, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The four fixed integers are β, K_0, M_C, M_F. Appendix A gives a closed finite recipe, including every seed character. All symbols below are polynomial abbreviations. Put

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & X &= wq,\\
k &= \eta + \zeta, & c &= kY + \eta, & \mu &= q^2 + 1,\\
A &= \mu Y(X+1) + 1, & \Delta &= A^2 - 1, & H &= 4A - 5,\\
V_0 &= X\mu^2 Y^2, & \ell &= V_0 k, & D &= \ell(\ell + k) + 1.
\end{aligned}$$

The source index, transport expression and index error are

$$\begin{aligned}
R &= [(q-1)(q-F) + \alpha](q^2-1) + (M_C + qM_F)J,\\
B_t &= (K_0 + w)C - F, \qquad & e &= k - R - 1.
\end{aligned}$$

The changed strong and auxiliary quantities are

$$\begin{aligned}
I_0 &= ic^2, & B_f &= 1 + \Delta D I_0^2, & Q &= \Delta^2 D I_0^2,\\
\delta &= Q - 1, & b &= c(T + \alpha) - R, & V &= B_f b - c,\\
C_y &= QV^2 - 1, & P &= AI_0 D, & W &= 1 + (q-1)b.
\end{aligned}$$

Finally define

$$\begin{aligned}
K_+ &= X(3c - 2X) + 2, & P_1 &= q^3(q-1), & M &= P_1\mu^2 H,\\
Z &= P_1\mu^2[H(n + V + P) + K_+]\\
&\quad + H[\mu^2((q-1)Y - q^3 W B_t) - P_1 e],\\
s &= \delta Z^2, & u &= \delta M^2 D B_f, & v &= M^2 C_y.
\end{aligned}$$

The requested single equation on this domain is

$$\boxed{\mathcal{P} = (s - u - v)^2 - 4uv = 0.} \tag{1}$$

In particular, this is an ordinary polynomial equation with fixed integer coefficients. The radicals and divisions used in its proof do not occur in its evaluation. The delivered arithmetic graph has 55 multiplications, 22 additions and 16 subtractions; every row, all ten inputs and all four coefficient ports are live.

The whole BFS-renamed compact44 coefficient tuple is used. It satisfies

$$\beta \text{ odd}, \quad \beta \geq 31, \quad K_0 > 0, \qquad 0 < M_C < \beta < M_F < 2\beta. \tag{2}$$

The values of β, M_C, M_F agree with the original compact44 tuple, but K_0 differs under the coherent state/color renaming. We use its separate source transfer, not a numerical identification of the two K_0 values [3]. The printed recipe has $\beta = 2^{590069} - 1$; the four payload bit lengths are $590069, 592884, 590069, 590070$ for β, K_0, M_C, M_F, respectively. The small operation count should not conceal these coefficient sizes.

Theorem 2.1. *For the complete fixed tuple in Appendix A, Equation (1) has a solution in $\mathbb{Z}_{>0}^{10}$ if and only if the calibrated source-61 system has a positive solution. By its inherited ordinary calibration, this is equivalent to the existence of a finite syntactic ZFC refutation. The exact total degree of $\mathcal{P}$ is 278.*

3 Two elementary extraction lemmas

Lemma 3.1 (Unequal two-root extraction)**.** *Let $r, y > 0$ have rational squares. If $r \neq y$ and a signed sum of r, y is rational, then both r and y are rational.*

Proof. Write $x = \varepsilon r + \eta y \in \mathbb{Q}$, with both signs in $\{1, -1\}$. Since $r \neq y$, $x \neq 0$. Then $2\varepsilon x r = x^2 + r^2 - y^2$, so r is rational, and the equation for x gives rationality of y. □

Lemma 3.2 (Adjacent-coefficient descent)**.** *Let $Q > 1$ and V be integers. If $y \in \mathbb{Q}$ satisfies $QV^2 - (Q-1)y^2 = 1$, then y is an integer.*

Proof. Replace V, y by their absolute values. Neither is zero. Write $y = a/m$ in lowest terms with $m > 0$. The equation implies $m^2 \mid Q - 1$. For $V = 1$ we have $y = 1$. For $V > 1$, set

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV.$$

The number V' is an integer and the same norm equation holds for (V', y'). Moreover $y > V$ and $QV > (Q-1)y$, by subtracting the relevant squares. Consequently $-V < V' < V$, and the norm excludes $V' = 0$. The reduced denominator of y' is still m, because $2Q - 1 \equiv 1 \pmod{m}$. Strong induction on $|V|$ reaches $V = 1$ and forces $m = 1$. □

Square-root saturation itself has a predecessor in the repository's nine-positive route [4]. The following use of it avoids the weighted three-root noncancellation argument of the ten-positive parent.

Proposition 3.3 (The saturated certificate)**.** *Every positive zero of (1) yields positive integers τ, f, y such that*

$$D = \tau^2, \qquad B_f = f^2, \qquad f^2 - \Delta(I_0\tau)^2 = 1, \qquad QV^2 - (Q-1)y^2 = 1.$$

In particular, $I = I_0\tau$ is a genuine strong Pell ordinate and $c^2 \mid I$.

Proof. At every positive input, $q \geq 32$, $A > 4$, $c \geq 3$, $I_0 \geq c^2$ and $D > 1$. Directly from the definitions,

$$Q - 1 > c^2 - 1 > 0, \qquad B_f > c + 2, \qquad B_f < Q - 1.$$

If $b = 0$, then $V = -c$; otherwise $|V| \geq B_f - c > 1$. Thus $C_y > 0$. Set $r = \sqrt{DB_f}$ and $y = \sqrt{C_y/\delta}$ as positive real roots. The four-conjugate identity for $(x^2 - a - b)^2 - 4ab$ gives

$$Z/M = \varepsilon r + \eta y. \tag{3}$$

Its prefactor $\delta^2 M^4$ is positive. If $r = y$, clearing denominators would give $(Q-1)DB_f = QV^2 - 1$. Reduction modulo D gives $0 = -1$, because $D \mid Q$, a contradiction.

Lemma 3.1 gives rational r, y. A rational number with integer square is an integer, so $r \in \mathbb{Z}$. Since $B_f \equiv 1 \pmod{D}$, $\gcd(D, B_f) = 1$. Prime valuations in the integer square $r^2 = DB_f$ show that both factors are squares. Lemma 3.2 gives $y \in \mathbb{Z}$. Only now define $I = I_0\tau$; its stated properties follow. □

4 Soundness: restoring the source in the right order

For $A \geq 2$, write

$$\chi_A(j) + \psi_A(j)\sqrt{A^2-1} = (A + \sqrt{A^2-1})^j.$$

The inherited normalized source proof supplies the strong-rank and signed auxiliary-index lemmas, and the immediate zero-offset parent supplies its ratio and projection refinements [2, 1]. Their exact hypotheses, rather than just their conclusions, are used below.

4.1 Folding and first growth before transport

The strong equation and auxiliary residues

$$V \equiv -c \pmod{f}, \qquad c \leq I,$$

give $c = \psi_A(p)$ for a positive odd index p. One way to see the folding step is to reduce an auxiliary Pell index modulo the strong period: $\psi_A(j+2m) \equiv -\psi_A(j) \pmod{\chi_A(m)}$ and $\psi_A(2m-j) \equiv \psi_A(j)$. The representatives have size at most I, whereas $f > 2I$. Thus the residue of c determines a positive representative. The odd auxiliary index preserves its parity through this folding.

The first norm gives $k = 2\psi_{2V_0+1}(n_0)$ for $n_0 \geq 1$. If $n_0 = 1$, then $k = 2$, $\eta = \zeta = 1$, and $c = 2Y + 1$ lies strictly between $\psi_A(1)$ and $\psi_A(2)$, which is impossible. Hence

$$c > kY \geq k \geq 4(2V_0 + 1) > 8X\mu^2 Y^2. \tag{4}$$

No transport conclusion, parity conclusion for q, or divisibility of e has yet been used. In particular, $c > 2X$ now proves $K_+ > 0$.

The unchanged mask estimate is $R < q^4 + (\alpha+2)q^2$. Equation (4) gives $R < c\alpha/4$ and hence $b > 0$. The same two cases $R \geq 0$ and $R < 0$ as in the parent give $|e| < b$, so $|e/\mu^2| < b/2$. Since $B_f < Q - 1$ and $V > 0$,

$$0 < y - V = \frac{V^2 - 1}{(Q-1)(y+V)} < \frac{b}{2}.$$

4.2 The upper center removes a large transport term

The new center has the exact defect

$$P^2 - r^2 = D(I_0^2 D - 1) > 0. \tag{5}$$

Thus $r < P$. Dividing the definition of Z by M in the proof gives

$$Z/M = n + V + P + K_+/H + Y/q^3 - WB_t/(q-1) - e/\mu^2.$$

Put $t = B_t/(q-1)$ and $\rho = e/\mu^2$ temporarily as rationals. The signed-root equation implies

$$n = \varepsilon r - P + (\eta y - V) - K_+/H - Y/q^3 + Wt + \rho. \tag{6}$$

If $B_t \leq -1$, then $Wt < -b$. The terms $\varepsilon r - P$, $-K_+/H$ and $-Y/q^3$ are negative; the other two errors together are less than b. Equation (6) would give $n < 0$. Consequently $B_t \geq 0$, and since $F > 0$ and $K_0 + w > 0$, also $C > 0$. The source masks now give

$$3q < R < q^4 - 2. \tag{7}$$

This is the important use of the upper center. The old weight contained a large strong-ordinate term. Here $W = 1 + (q-1)b$ suffices. Its degree is 5, which will be used in the exact degree calculation.

4.3 Rank, parity, projection, and the unscaled source

For reference, the normalized rank argument is short. If $c = \psi_A(p) > 2$ and $c^2 \mid \psi_A(m)$, strong divisibility gives $m = pv$. Put $D_m = \chi_A(p)$. Then $\psi_A(m) = c\psi_{D_m}(v)$ and $\psi_{D_m}(v) \equiv vD_m^{v-1} \pmod{c}$, with D_m a unit modulo c. Thus $c \mid v$, so $pc \mid m$ and $m > 2p$.

Apply this to the recovered $I = I_0\tau$. The signed auxiliary-index lemma also uses $V \equiv -R \pmod{c}$, which follows from $f^2 \equiv 1 \pmod{c}$. Its strict bounds $0 < p, R < c/2$ follow from (4), (7), and Pell growth. It therefore gives $p = R$. Since p is odd, R is odd. Odd q would make J even and the defining expression for R even, because β is odd. Thus q is even.

Only at this point use $\gcd(\mu^2, q^3(q-1)H) = 1$. In the signed-center identity, n, V, P, r, y are integers, so a partial-fraction argument yields $\mu^2 \mid e$. The first-Pell congruence and the comparison $2V_0 + 1 > A$ now force $n_0 = (R+1)/2$. Write $R = 2h+1$; then $h \geq 48$. For $a = Y(X+1)$, the zero-offset ratio estimate is

$$\xi < \frac{\psi_A(2h+1)}{\psi_{2V_0+1}(h+1)} < \xi(1+1/a)^{2h}, \qquad \xi = \frac{(X+1)^{2h}}{X^h}.$$

It gives $Y > X^h/3$ and $X^{h+1} < H$.

Set $K_{\mathrm{quad}} = X(X + 2(A-2)c) - 1$. The exact identity $K_+ = HXc - 2K_{\mathrm{quad}}$ and the oddness of H transfer the projection divisibility to K_{quad}. The Pell projection then yields

$$H \mid q^3(q-1)(X - 2^R)(2^R X + 1).$$

For $X \geq q \geq 32$ and $h \geq 48$, its entire absolute value is less than

$$q^4 2^{2R+2} X^2 < X^{h+1} < H.$$

The dividend is therefore zero, and its second factor is positive, so $X = 2^R$. No nonunit factor is cancelled through the composite modulus. The remaining partial fractions give $q^3 \mid Y$ and $q - 1 \mid B_t$. The retained fixed-minus parity lemma gives $R \equiv 3 \pmod 4$, and ratio rounding recovers the required half-binomial coefficient Y.

These are exactly the data from which the parent constructs fresh unscaled source-61 witnesses. The reconstruction does not divide the scaled base or reuse the current strong witness as an unscaled one. Its calibrated source theorem, including the allowed positive $J = 1$ case, yields a finite refutation. This proves soundness.

5 Completeness: the new witnesses really exist

It is not enough to retain the predecessor's witnesses: the new strong ordinate must be divisible by τ, and the upper center subtracts a large positive term from the expression for n. Both points need proof.

Canonical source data give $X = 2^R$, $R \equiv 3 \pmod 4$, $q^3 \mid Y$, and $B_t = (q-1)t$ with $t \geq 0$. Retain J, F, α, w, Y, and choose

$$c = \psi_A(R), \quad k = 2\psi_{2V_0+1}((R+1)/2), \quad \tau = \chi_{2V_0+1}((R+1)/2).$$

The unchanged ratio argument gives $Y < c/k < Y + 1$, so $\eta = c - kY$ and $\zeta = k - \eta$ are positive integers. Also $\rho = (k - R - 1)/\mu^2$ is a positive integer.

Lemma 5.1 (Divisible strong witnesses)**.** *For any positive integer L and $A \geq 2$, there is a positive index m with $L \mid \psi_A(m)$.*

Proof. The matrix $\begin{pmatrix} A & A^2-1 \\ 1 & A \end{pmatrix}$ has determinant one. Its reduction modulo L belongs to a finite group, so some positive power is the identity. The lower-left entry of its mth power is $\psi_A(m)$. □

Apply the lemma with $L = c^2\tau$ and set

$$I = \psi_A(m), \qquad f = \chi_A(m), \qquad i = I/(c^2\tau), \qquad I_0 = I/\tau.$$

These are positive integers, $I_0 = ic^2$, and the saturated strong equation holds. No index variable or exponent is introduced into the polynomial.

Put $S = \Delta I$ and $Q = S^2$. The standard odd-index auxiliary witnesses are

$$V_* = \frac{\chi_S(R)}{S}, \qquad y_* = \psi_S(R), \qquad T_* = \frac{V_* + c + Rf^2}{cf^2}.$$

The retained odd-index polynomial identities give $V_* \equiv -c \pmod{f^2}$ and $V_* \equiv -R \pmod{c}$. Since $f^2 \equiv 1 \pmod{c}$, T_* is an integer and $V_* = f^2(cT_* - R) - c$.

Here $R \geq 5$. Monotonicity and the explicit fourth and fifth Pell terms give

$$V_* \geq \chi_S(5)/S = 16Q^2 - 20Q + 5 > Q^2,$$
$$u_* := y_* - V_* = \frac{\psi_S(R-1)}{S} \geq 8Q - 4.$$

The exact gap identity follows from $\chi_S(R) - S\psi_S(R) = -\psi_S(R-1)$. Since $f^2 < Q$ and $Q/c > q > \alpha$, $T_* > \alpha$. Thus $T = T_* - \alpha$ is a positive integer and the equation's V equals V_*.

The projection gives the positive integer $\kappa = K_+/H < Xc$. Set $s_0 = Y/q^3 \geq 1$ and define

$$n = Wt + \rho + \tau f + u_* - P - s_0 - \kappa. \tag{8}$$

It is an integer. Moreover $P < Q$ because $Q/P = \Delta^2 I_0/A > 1$. Equation (4) implies $X < c/8$ and $Y < c$, hence $s_0 + \kappa < Xc + Y < c^2 < Q$. Therefore

$$P + s_0 + \kappa < 2Q < 8Q - 4 \leq u_*.$$

All other terms in (8) are nonnegative, so $n > 0$. The normalized center now equals $\tau f + y_*$, making one factor of (1) vanish. This proves completeness, including positivity of both changed witnesses i and n.

6 Exact degree and an unrestricted-integer version

The direct syntactic degree propagation of the 93-gate graph gives 316. The exact degree is smaller. Introduce proof-only polynomial abbreviations

$$z = Z - M(V + P), \qquad d = P^2 - DB_f = (I_0 D)^2 - D,$$

and

$$B = 2M(P+V)z + z^2 + M^2 d, \qquad C_0 = \delta B - M^2(V^2 - 1).$$

A complete ring identity is

$$\begin{aligned} \mathcal{P} = {} & C_0^2 + 4\delta M^2 PVC_0 - 4\delta M^4 P^2(V^2-1) \\ & + 4\delta M^4 d(QV^2 - 1). \end{aligned} \tag{9}$$

It follows by substituting $Z = M(V + P) + z$ and $DB_f = P^2 - d$ into the norm and using $\delta = Q - 1$; the checker compares every coefficient of this six-indeterminate identity.

The relevant degrees are

$$\deg Q = 48, \quad \deg M = 13, \quad \deg V = 42, \quad \deg P = 28, \quad \deg z \le 19, \quad \deg d = 46.$$

Thus $\deg B \le 74$ and $\deg C_0 \le 122$. The four terms in (9) have degrees at most $244, 266, 240, 278$. The only degree-278 contribution is $4Q_0^2M_0^4d_0V_0^2$, where subscripts denote leading homogeneous forms. To avoid confusing V_0 here with the earlier register, its fully expanded factorization is

$$2^{10}q_0^{146}i^{10}w^{28}Y^{60}(\eta + \zeta)^{32}(q_0 - F)^2, \qquad q_0 = \beta J. \tag{10}$$

In particular the coefficient of $J^{146}i^{10}w^{28}Y^{60}\eta^{32}F^2$ is $2^{10}\beta^{146} > 0$. The exponents sum to 278, proving the claimed exact degree. This proof is independent of the numerical values of K_0, M_C, M_F.

For an immediate unrestricted-integer equation, substitute, for each of the ten positive inputs,

$$x_j = 1 + a_j^2 + b_j^2 + c_j(c_j + 1), \qquad a_j, b_j, c_j \in \mathbb{Z}. \tag{11}$$

This expression parametrizes precisely the positive integers. The converse uses the three-square theorem on $4(x_j - 1) + 1$, with one odd and two even squares. Thus the resulting 30-variable equation has the same refutation equivalence. Its degree is exactly 556: substitution of independent nonzero quadratic leading forms into the nonzero product (10) stays nonzero. The stated evaluator adds seven operations per input, for $93 + 70 = 163$ operations. This companion is explicit and useful for domain clarity, but it is not advertised as a new unrestricted variable-count or low-degree record.

7 A direct consistency quartic in 576 unrestricted integers

The direct proof-certificate route admits a new simplification: a quadratic bound on a proposed power output can replace its sign predicate. Once the indexed Pell coordinates are oriented, a small signed residue identifies the output and recovers its sign. Combining this with two local source identities and a factorial digit obstruction gives the following explicit quartics.

Equation	Variables	Quadratic rows	Terms	Height
Previous principal	604	372	9,676	2,445,790,974,704
New principal	**576**	368	**7,172**	2,445,790,974,704
Previous lower height	609	377	8,823	964,213
New lower height	**582**	374	**6,200**	964,213

All four total degrees are exactly four. The two new equations are literally

$$P_{576}(x) = \sum_{j=0}^{367} R_j(x)^2 = 0, \qquad P_{582}(x) = \sum_{j=0}^{373} \widetilde{R}_j(x)^2 = 0,$$

where the accompanying JSON files list every quadratic residual and every nonzero expanded coefficient. There are no coefficient ports or unmaterialized variable powers. The first equation saves 28 variables and 2,504 monomials at unchanged height; the second saves 27 variables and 2,623 monomials. These are measured improvements, not a minimality claim. This direct C222 route does not use K215, Jones compression or Sun's integer reduction.

The precise new theorem is integer-zero existence equivalence with the pinned 75-predicate direct-C222 source. Its existing ordinary refutation/consistency calibration transfers separately. All arguments here are finite integer arithmetic that can be performed in ZFC; no consistency or external soundness assumption is made. A new Lean, Rocq or sealed object-language ZFC proof object is not claimed.

7.1 The exact source change

The source is pinned at ProveIt commit `84e16e43092614bae262ca59651cb9d77e31a255`, in the `factorial-radius-output-sign-profile/raw` research packet. Its body has 1,259 coordinates and 1,177 quadratic residuals. A predicate $E \geq 0$ is represented by

$$E - a^2 - b^2 - c(c+1) = 0$$

with three private integer variables. These witnesses exist exactly when $E \geq 0$, by the three-square theorem applied to $4E + 1$.

Delete exactly the twelve scalar predicates with raw indices

$$6, 104, 106, 575, 584, 592, 600, 609, 618, 627, 636, 825.$$

Keep the other 63 predicates. Put

$$S = \{6, 106, 573, 583, 591, 599, 607, 617, 625, 635\}.$$

Change only body residual 1176, originally $x_{643} - x_{1258}$, to

$$x_{643} - x_{1258} + \sum_{i \in S} (x_i - x_i^2).$$

This adds no coordinate and preserves quadratic degree. Every other body residual is unchanged. The resulting raw equation has 1,448 integer coordinates and 1,240 quadratic residuals.

Let $\mathcal{A}$ be the set of all 43 distinct base, exponent and output indices in the 21 actual power calls. The original frame gate identities now give exactly

$$W = 2 + \sum_{i \in \mathcal{A} \setminus S} x_i + \sum_{i \in S} x_i^2, \qquad W = x_{643}. \tag{12}$$

The complete list $\mathcal{A}$ and the actual source-row ideal are included in the certificate. A squared contribution bounds either sign of an integer because $z^2 \geq |z|$.

7.2 Signs and a dominating Pell frame

Write $f = x_{87}$, $r = x_{94}$, $s = x_{95}$ and $N = x_{93}$. The retained predicates give $f \geq 2$, $r, s \geq 0$. Exact source rows 78–85 imply

$$N = fr + (f-1)s + f \geq 2.$$

Separately, an eleven-row polynomial identity gives

$$f(x_{104} + 1) = f x_{59} + x_{131} + 3. \tag{13}$$

The two right-hand coordinates retain direct nonnegativity predicates. Thus $x_{104} + 1 > 0$ and $x_{104} \geq 0$, without any power interpretation. The full eleven integral polynomial multipliers are supplied.

Filtering the old prefix proof now proves 604 of the 643 semantic coordinate signs. Its only postponed special bound concerns x_{130}. The only unknown-sign frame arguments are precisely the ten members of S. All actual bases and seed exponents are already positive. In particular, with $t = x_4$, $K = x_{86}$ and $f_1 = x_{107}$,

$$t \geq 1, \quad K \geq 3, \quad f_1 \geq 0,$$

$$M = x_{129} = 1 + f(x_{59} + 1) \geq 3,$$
$$b = x_{530} = M(1 + N + x_{113}) \geq 9, \qquad x_{531} = b + 1 \geq 10,$$
$$x_{103} = 1 + (t + 1)f \geq 5.$$

Consequently the new frame gives, for every actual call (n, k, m),

$$W \geq 3, \qquad 1 \leq n, k < W, \qquad |m| < W. \tag{14}$$

The retained initial modulus and its identity $T_0 = 2A(t + 1) - (t + 1)^2 - 1$ force $A = x_{644} \geq 2$. The unchanged bootstrap norm is

$$A^2 - 1 = W^3(W + 2)Z^2.$$

Here $Z \neq 0$. At Pell base $W + 1$ this is the solution $(A, W|Z|)$. Classification and the recurrence modulo W imply that its positive index is a multiple of W. Hence

$$A > W^W, \qquad T_n = 2An - n^2 - 1 \geq A > W^W. \tag{15}$$

The last inequality follows from the exact decomposition

$$T_n - A = (2n - 1)(A - n - 1) + (n - 1)(n + 2).$$

The retained $V \geq 1$ predicate and the unchanged product identity $V = (\prod_i Y_i^2)(v + 1)$ make every seed ordinate nonzero and give $Y_i^2 \mid V$. The common norm implies $|U| \geq A$. From $B = 1 + 4V(q + 1)$ and $B \equiv A \pmod{U}$, the case $q = -1$ is impossible: it would force $U \mid A - 1$, although $0 < A - 1 < |U|$. Every other q gives $|B| \geq 3$. Thus the pinned signed-base index theorem is available whenever the particular short-interval condition $k \leq |Y|$ has been established.

7.3 A power output may initially be signed

Signed-output lemma. In this section $\chi_k(a), \psi_k(a)$ use an index subscript: they start at $(1, 0)$ and satisfy $\chi_{k+1} = a\chi_k + (a^2 - 1)\psi_k$, $\psi_{k+1} = \chi_k + a\psi_k$. Assume (14)–(15), $n \geq 2$, and

$$|X| = \chi_k(A), \quad Y = \psi_k(A) > 0, \qquad X - (A - n)Y - m = T_n h,$$

where m, h are unrestricted integers. Then $X = \chi_k(A)$ and $m = n^k$.

Proof. Put $p = \chi_k - (A - n)\psi_k$ and $q = \chi_k + (A - n)\psi_k$. The Pell recurrence and norm give $p \equiv n^k \pmod{T_n}$ and $pq = 1 + T_n\psi_k^2$. If $X < 0$, then $T_n \mid mn^k + 1$. This dividend is nonzero for every integer m, since $n^k \geq 2$, but

$$|mn^k + 1| \leq (W - 1)^W + 1 < W^W < T_n.$$

This is impossible. Hence $X > 0$, and $T_n \mid m - n^k$. Now

$$|m - n^k| \leq W - 1 + (W - 1)^{W-1} < W^W < T_n,$$

so $m = n^k$. Notice that positivity of Y is essential: if both main coordinates could be negative, $m = -n^k$ would survive. Every application below first supplies that positive ordinate.

7.4 Restoring the deleted predicates in order

The eight range quotients. For actual calls $10, 12, 13, 14, 15, 17, 18, 20$, all eight original $Y \geq k$ predicates remain. The respective outputs are $573, 583, 591, 599, 607, 617, 625, 635$; all bases are $x_{531} \geq 10$. The signed index theorem followed by the signed-output lemma makes these outputs their intended positive powers. Their exact range rows are

$$m = (Me)q + r, \qquad e = r + s + 1, \qquad r, s \geq 0.$$

Here $M = x_{129}$, not the radix $b = x_{530}$. Since $0 \leq r < e \leq Me$, a negative q would make $m < 0$. All eight quotients and their 24 dependent signs are recovered. Only the seven prefix signs

$$6, 106, 124, 125, 130, 132, 133$$

remain unproved.

The first factorial. None of those seven coordinates is a sign premise of the pinned local first-factorial argument. Put $d = fK^2$ and $C = x_{91}$. Its exact inputs are

$$C = r + s + 1, \quad fC = N + s, \qquad x_{532} = x_{533}(C + ba) + u,$$

$$x_{533} = u + v + 1, \qquad b = C + z + 1,$$

with $a, u, v, z \geq 0$, together with actual calls $1, 5, 6$. All three outputs N, x_{532}, x_{533} remain known nonnegative.

For either missing first-factorial index, $|Y| \geq 2$ implies $|Y| \geq 2A > W > k$, so the index theorem and a nonnegative-output supporting call recover its true positive pair. If $|Y| = 1$, the four possible residues $n, -2A + n, 2A - n, -n$ and the small-output bound leave exactly $X = A, Y = 1, m = n$. Thus the effective indices are $e \in \{K, 1\}$ and $j \in \{K + d, 1\}$, with

$$N = (d + 1)^e, \quad x_{533} = b^e, \quad x_{532} = (b + 1)^j.$$

If $j = 1, e = K$, the digit equation contradicts $C \geq 1$; if $j = e = 1$, it forces $C = 1$ and then $f \geq d+1 > f$. Therefore $j = K + d$. If $e = 1$, the bound $b \geq (f + 1)(d + 2) > K + d$ identifies the first binomial digit as $C = K + d$, and the ceiling equation forces $f = 1$, again impossible. Both index guards are restored. The pinned local factorial lemma now gives $f = K!$, so f is even. Its complete low-digit and self-referential-radius proof is bundled; it uses these three calls, not an interpretation of the remaining powers.

Orienting the t-pair with a different output. The base $x_{103} = 1 + (t + 1)f$ is odd. The t-seed norm and its actual call-2 residue modulo two imply x_{104} is odd, without interpreting t. Hence t and $k_{122} = t + x_{104} + 2$ have opposite parity. An even index excludes $|Y| = 1$: the signed abscissa half-period argument makes the comparison index odd, while its ordinate congruence modulo four makes it even. Thus its true index is recovered, and a positive-output call orients its pair.

If t is even, use *call 2*, whose output $x_{104} \geq 0$ was proved in (13); its base is at least five. If k_{122} is even, use *call 9*, with output $x_{560} \geq 0$. The retained product of these two index slacks then makes the other slack nonnegative. The indexed t-pair is oriented by call 2 also when t is odd. This change of supporting call avoids assuming the missing sign of call 0's output x_6.

With that positive t-pair, the signed-output lemma at bases $t + 1$ and f recovers $x_6 = (t + 1)^t > 0$ and $x_{106} = f^t > 0$. The remaining source rows give

$$x_{125} = x_{123}x_{107}x_{106} \geq 0, \quad x_{125} = M + x_{126}x_{130}, \quad x_{126} = M + x_{131} + 1 > M.$$

Consequently $x_{130} \geq 0$; its dependent signs follow. The entire 643-coordinate semantic prefix is now natural.

The second factorial's shifted index. It remains to recover the deleted predicate at 825. With t exact, actual calls 4 and 8 give

$$N_1 = x_{113} = (1 + f_1 t^2)^t \geq 1, \qquad x_{547} = b^t.$$

The second factorial rows give $C_1 = x_{114} + x_{115} + 1 \geq 1$ and $f_1 C_1 = N_1 + x_{115}$, so $f_1 \geq 1$. Its shifted seed has index $t + f_1 t^2$. The large-ordinate case recovers its true positive pair as above. In the small case $|Y| = 1$, the signed residue analysis forces $x_{546} = b + 1$. Its exact digit equation is

$$x_{546} = b^t(C_1 + ba_1) + u_1, \qquad C_1 \geq 1, \quad a_1 \geq 0, \quad 0 \leq u_1 < b^t.$$

For $t \geq 2$ the right side is at least $b^t > b + 1$. For $t = 1$, division forces $C_1 = 1$, whereas $f_1 C_1 = f_1 + 1 + x_{115} > f_1$. Both cases are impossible. Thus the omitted shifted-index slack is strictly positive.

7.5 Completeness requires a fresh private frame

All 75 parent predicate values have now been recovered. The unchanged Pell and small-residue arguments give every true power value; they use only the dominating bounds, not the literal old linear frame. Nevertheless, adjoining guard triples alone does *not* produce an old source zero: its original frame equality usually fails.

Hold the natural semantic prefix fixed and reconstruct the private Pell data. Choose either its old linear frame W_0 or the new squared one, and set

$$A_0 = \chi_{W_0}(W_0 + 1), \qquad Z_0 = \psi_{W_0}(W_0 + 1)/W_0.$$

The recurrence modulo W_0 proves the division is exact. Choose all main pairs at their actual indices. In the finite quadratic ring modulo $L = 4\prod_i Y_i^2$, the unit $A_0 + \sqrt{A_0^2 - 1}$ has finite order. This supplies a positive common Pell solution with $U \equiv 1$, $V \equiv 0 \pmod{L}$. The norm gives $\gcd(U, V) = 1$, and U is odd. CRT supplies $B \equiv A_0 \pmod{U}$, $B \equiv 1 \pmod{4V}$, with $B > A_0$. Take the comparison pairs at their true indices and use their integral progression quotients. The product quotient is $V/(\prod_i Y_i^2) - 1 \geq 0$.

For each power call, its integral quotient follows from

$$H_0 = 1, \quad H_{j+1} = nH_j + T_n\psi_j(A_0), \quad H_j = \chi_j(A_0) - (A_0 - n)\psi_j(A_0).$$

Reevaluate the derived t+1 pair, discriminant, moduli and private arithmetic gates. Every one of the 109 private nongate residuals is the sole nongate, with coefficient ± 1, in an exact row identity for a target just satisfied. All other terms are unit gate equations. The independent audit checks this entire coverage, including the 474 private topological gate definitions. Fresh three-square witnesses satisfy every private guard.

Using the old linear frame constructs an old 75-predicate zero from any new zero. Conversely, use the old source's true powers and the same completion with the squared frame, omitting the twelve predicates. This constructs a new zero. The maps establish existence equivalence; they are not asserted to be inverse on private witnesses.

7.6 Exact algebraic verification

The final projections contain respectively 872 and 866 monic eliminations. Each deleted coordinate has a unique polynomial integer restoration, every surviving residual stays quadratic, and no unconstrained variable is discarded. Reversing the sequence gives a bijection of complete integer zero sets, including guard rows. An independently authored checker verifies the source changes, 604-sign initial dependency boundary, all staged interfaces and all 109 private completion targets. A separate checker imports no optimizer, replays every signed restoration and collects every quartic coefficient. Both normal and optimized Python modes pass, and changed source rows, guards, inverses and coefficients are rejected.

These finite checks establish the literal polynomials and their algebraic transformations. The unbounded arguments are the Pell, integer-order, local factorial and finite CRT proofs given above and in the bundled pinned sources. They do not prove Con(ZFC) itself. The next concrete research questions are to minimize the outputs needed to orient all seed pairs, and to optimize the signed frame together with degree-preserving monic projection: a square is free in the raw coordinate count but may prevent a later elimination.

8 Verification, integration, and limitations

The mathematical specification and reproducibility package are complementary. The ten-positive polynomial has a complete closed coefficient recipe in this report and a literal arithmetic graph in `ten_positive/equation.json`. The direct quartics have complete residual systems, complete collected monomial tables and all integer reconstruction maps in `direct/`. No unknown coefficient or program index remains to be chosen later.

The checks performed for the ten-positive result include:

- the entire retained/changed boundary against the 99-gate parent;
- exact new front-register identities and every coefficient of the four-conjugate norm and double-center identity;
- degree bounds for every centered summand, the complete nonzero leading form, and its explicit degree-278 coefficient;
- all four coefficient payloads reconstructed from the 436-character seed, including the 943 tiles and 66,062 finite native convolution checks;
- separate independent proof review of root extraction, dependency order, finite-period witnesses and the auxiliary growth needed for $n > 0$.

The direct route additionally checks every original-row identity used for the changed bound and recovered predicates, every monic integer inverse, every residual, and every coefficient of both expanded quartics. The normal and optimized Python replays preserve the checks: essential verifications do not disappear under `python -O`. Bounded examples and modular samples are diagnostic controls, not substitutes for the unbounded proofs.

The archive includes a top-level `verify.py`, a source ledger, licenses and deterministic build commands. The original repository is not required to evaluate the shipped polynomials or run the standalone checks. The source pin remains useful for reviewing the inherited semantic calibrations.

There are two principal limits. First, the ten-positive route still uses the native K215 source calibration; its smaller arithmetic does not make it an independent foundational encoding. The direct quartic route supplies that different encoding. Second, neither new ordinary equivalence is presented as a new end-to-end proof-assistant theorem. Exact algebra checks certify the finite artifacts and support transcription accuracy; they do not themselves certify all source semantics or prove Con(ZFC).

9 Further research questions

1. **Can the signed-output frame be generalized without creating expensive quadratic dependencies?** Squaring an argument removes its sign premise but may prevent an earlier monic quadratic substitution. The actual variable count must be measured after projection. A joint choice of frame terms and elimination order may improve the 576-variable profile.
2. **Can the two-root adjacent certificate eliminate another positive unknown?** The present reduction compresses square conditions, not the free source coordinates or the auxiliary index witness. A further reduction must supply complete source reconstruction and canonical completeness; a shorter norm formula alone is insufficient.
3. **Can better rational centers lower the degree again?** The upper center $P = AI_0D$ has a degree-46 defect. A closer center may reduce that defect, but clearing a new denominator can increase degree and interfere with the projection's coprimality. Both effects must be counted together.
4. **Can saturation transfer to the signed low-variable branch?** An unrestricted-integer witness absorbs changes to the center easily, but some signed parents require an *odd* strong multiplier. A finite-period argument guarantees divisibility, not oddness. For example, at Pell base 2, the least index with $7 \mid \psi_2(m)$ is 4; all such indices are multiples of 4 and their ordinates are even. An odd-index divisibility theorem is a real additional obligation, not an automatic corollary of this packet.
5. **Can the ordinary source calibrations be internalized with small certificates?** The local root extraction and finite-period lemmas are natural targets once formalization resumes. The final internal-ZFC statement must still quantify over all internally coded witnesses.

A A closed recipe for every coefficient

This appendix retains the entire coherent BFS-renamed compact44 recipe from the pinned one-page source [3]. Its K is the main equation's K_0. All summation ranges, indices, sets, ranks, quotients and exponents here are fixed or bound in the coefficient definitions; none is an extra Diophantine unknown. The recipe was reconstructed and compared with every raw payload byte. It is attributed as inherited data, not as a new machine construction. In particular, the initializer uses the decoded first target.

A.1 The four fixed integers

Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805},\end{aligned}$$

$$K = \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),$$

where

$$\Phi(x) = (1 + 2^{51})G_{943}(x) - 2^{10} \sum_{t \in \mathcal{T}} v(t) x^{\varrho(t)} + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x),$$

$$\mathcal{C} = \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.$$

The rational intermediate in K cancels exactly; all four results are positive integers.

A.2 The 943 fixed tiles and their ranks

Use the actions u_j, v_j, d_j decoded below. Let

$$h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\},$$
$$\mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\},$$
$$r_q = 6 + |\mathcal{R} \cap \{0, \dots, q-1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q-1\}|.$$

An interval ending at -1 is empty. The following union is $\mathcal{T}$; the index ranges in each row are fixed:

range	tile (w, n, e, s)
$j = 2q + b < 430$	$((1 - d_j) l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$
$b = 0, 1$	$(0, h(215, b), 0, h(215, b)),\ (0, b, 0, b)$
$q \in \mathcal{R},\ b = 0, 1$	$(r_q, b, 0, h(q, b))$
$q \in \mathcal{L},\ b = 0, 1$	$(0, b, l_q, h(q, b))$
$t \in \{0, 1, 432, 433\}$	$(0, t, 0, 438)$
	$(2, 434, 2, 0),\ (2, 435, 3, 1)$
	$(3, 436, 4, h(u_0, 0)),\ (4, 436, 4, 0),\ (5, 0, 0, 0)$
$t = 434, 435, 436$	$(1, 438, 1, t)$
$a \in \{0, 1, 4\},\ b = 0, 1, 2$	$(a, 437, b, 437)$

Put $s_* = (2, 435, 3, 1)$, $f_* = (5, 0, 0, 0)$, and $\mathcal{T}' = \mathcal{T} \setminus \{s_*, f_*\}$. For $t = (w, n, e, s)$ set

$$\kappa(t) = 2^{30} w + 2^{20} n + 2^{10} e + s,$$
$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34},$$
$$\varrho(s_*) = 0, \quad \varrho(f_*) = 1,$$
$$\varrho(t) = 2 + \sum_{z \in \mathcal{T}'} \left\lfloor \frac{2^{40} + \kappa(t) - \kappa(z) - 1}{2^{40}} \right\rfloor \quad (t \in \mathcal{T}').$$

A.3 A closed decoder for the 430 actions

Put $D_b(A,k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

A.4 The complete seed numeral

The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSD0Tgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

B File formats and reproduction

An arithmetic graph row `[name,op,left,right]` defines one new register using one of `+,-,*`; an operand is either an earlier name or an integer literal. The output register is `N`. Coefficients are unsigned little-endian integers stored in deterministic compressed payloads; the closed recipe above also constructs them without reading those payloads.

The direct polynomial files use explicit sparse monomial lists. A monomial's variable indices and exponents, together with its integer coefficient, determine it completely. The residual systems are convenient for proof and evaluation, while the expanded files permit independent coefficient-level comparison. The direct package README specifies the exact JSON fields.

From the unpacked archive directory:

```
python3 -B verify.py
python3 -B -O verify.py
python3 -B ten_positive/build.py --check
python3 -B ten_positive/rebuild_coefficients.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The first two commands replay the shipped finite certificates. The report does not require network access or a proof-assistant installation to build. The full source ledgers distinguish copied upstream results from new files.

References

[1] ProveIt contributors, *Ten strictly positive integers, 99 operations, degree 368*, `Computability/DiophantineConsistency/candidates/ten-positive-zero-offset/`, at the fixed snapshot. https://github.com/VladimirReshetnikov/ProveIt/tree/84e16e43092614bae262ca59651cb9d77e31a255/Computability/DiophantineConsistency/candidates/ten-positive-zero-offset.

[2] ProveIt research package 61, *Congruence carriers*, `con-zfc-diophantine-forms/61-congruence-carriers/report/congruence_carriers.tex`, especially the source theorem, normalized rank, signed auxiliary index and positive auxiliary reconstruction. The complete source is copied in `ten_positive/upstream/SOURCE61.tex`.

[3] ProveIt contributors, *A complete ten-variable consistency equation*, `candidates/one-page-short10/`, and the inherited complete BFS source transfer. https://github.com/VladimirReshetnikov/ProveIt/tree/84e16e43092614bae262ca59651cb9d77e31a255/Computability/DiophantineConsistency/candidates/one-page-short10.

[4] ProveIt contributors, *Tightened nine-positive main-root folding*, `research/nine-folded-saturation-tight/PROOF.md`, at the fixed snapshot. The source proof is copied as `ten_positive/upstream/NINE_SATURATION_PRIOR.md`.

[5] ProveIt contributors, *Fresh 75-predicate integer quartic profiles*, `research/factorial-radius-output-sign-profile/`, at the fixed snapshot. https://github.com/VladimirReshetnikov/ProveIt/tree/84e16e43092614bae262ca59651cb9d77e31a255/Computability/DiophantineConsistency/research/factorial-radius-output-sign-profile.

[6] ProveIt contributors, direct C222 source, shared powers and shared index certificates; signed comparison-base, paired-index, factorial-radius and output-sign reductions. Exact source paths and hashes are included in `direct/source/` and the direct source ledger.

[7] Zhi-Wei Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2 September 2026. https://arxiv.org/abs/2609.01594.